

Bambu Lab LED Status Bar

(DIY version using Home Assistant + ESP32 + WLED or MQTT)

There's a ready-to-use progress bar kit available from BIQU if you prefer something plug-and-play:

[BIQU Panda Status LED Light Bar Kit H2D-Style](#)

I wanted to build my own version instead, mostly because I already had the parts and enjoy putting things together.

I followed this video and adapted the approach to work on my P2S:

<https://www.youtube.com/watch?v=SImwERgli7w>

The goal:

Use an **ESP32** to control a **5V WS2811 COB RGB LED strip** and turn it into a visual printer status and progress bar.

The current version uses **WLED + Home Assistant + the Bambu Lab integration** to monitor the printer and update the LED strip automatically during heating, calibration, printing, errors and other printer states.

Project Update 24/09/26 – Home Assistant V2 + Standalone MQTT Version

After receiving feedback and suggestions from the community, I made several corrections and improvements to the project.

The **Home Assistant configuration has now been updated to V2**, with better printer-state handling, improved HMS / print error recovery, cleaner WLED control and a dedicated filament status.

I also finally spent some time researching the Bambu Lab MQTT communication in more detail, and now a **standalone MQTT version** is available.

The standalone MQTT version will allow the ESP32 to communicate directly with the printer without requiring Home Assistant.

Parts I used / Shopping List:

- [ESP32 WROOM](#) - runs WLED / custom firmware (MQTT)
- [24V → 5V buck converter \(3A\)](#) - the Bambu PSU is 24V, LEDs + ESP32 need 5V ([option 2](#) adjustable) - *Only required if you want to power the ESP32 and LED strip from the printer's internal 24V supply.*
- [WS2811 COB RGB LED strip](#) - Smooth continuous RGB light (I had the 5v 720, and used a 25cm piece / 10 sections/leds)
- [470Ω resistor](#) - goes inline on the data wire to clean the signal and avoid flicker
- [1000µF 16V capacitor](#) - stabilizes the 5V line
- **3M double-sided adhesive tape** — used to stick the LED strip along the front edge of the print-bed frame

- **Wires** — to connect everything

Before You Start

If you're planning to build this, I really recommend watching the videos linked in this guide, especially for:

- opening the printer safely without guessing
- understanding the internal wiring layout
- configuring WLED (*not required on MQTT version*)
- setting up Home Assistant and the automation (*not required on MQTT version*)
- flash the WLED or MQTT firmware first before wiring up

They helped me a lot while building my version and will probably save you some time too.

Power Options

There are two ways to power the ESP32 and LED strip, depending on how you prefer to install the project.

Option 1 – USB Power (Easiest)

If you don't want to open your printer or connect anything to its internal power supply, you can power the ESP32 and LED strip using an external 5V USB power supply.

For my 25cm LED strip with 10 controllable sections, a suitable USB power supply can power both the ESP32 and LEDs.

This works with both the Home Assistant and standalone MQTT versions. The ESP32 communicates with the printer over Wi-Fi, so no physical connection to the printer is required.

Option 2 – Internal Printer Power

For a cleaner installation without an external USB cable, you can use the printer's internal 24V power supply with a 24V-to-5V buck converter.

This was the approach I used for my original build.

Make sure the printer is completely powered off and unplugged before accessing its internal components. If you're not comfortable working around a power supply, use the external USB option instead.

Important: If using USB power, make sure your power supply can handle the combined current of the ESP32 and LED strip. Longer strips may need a separate 5V supply. When using separate supplies, the ESP32 and LED strip must share a common ground.

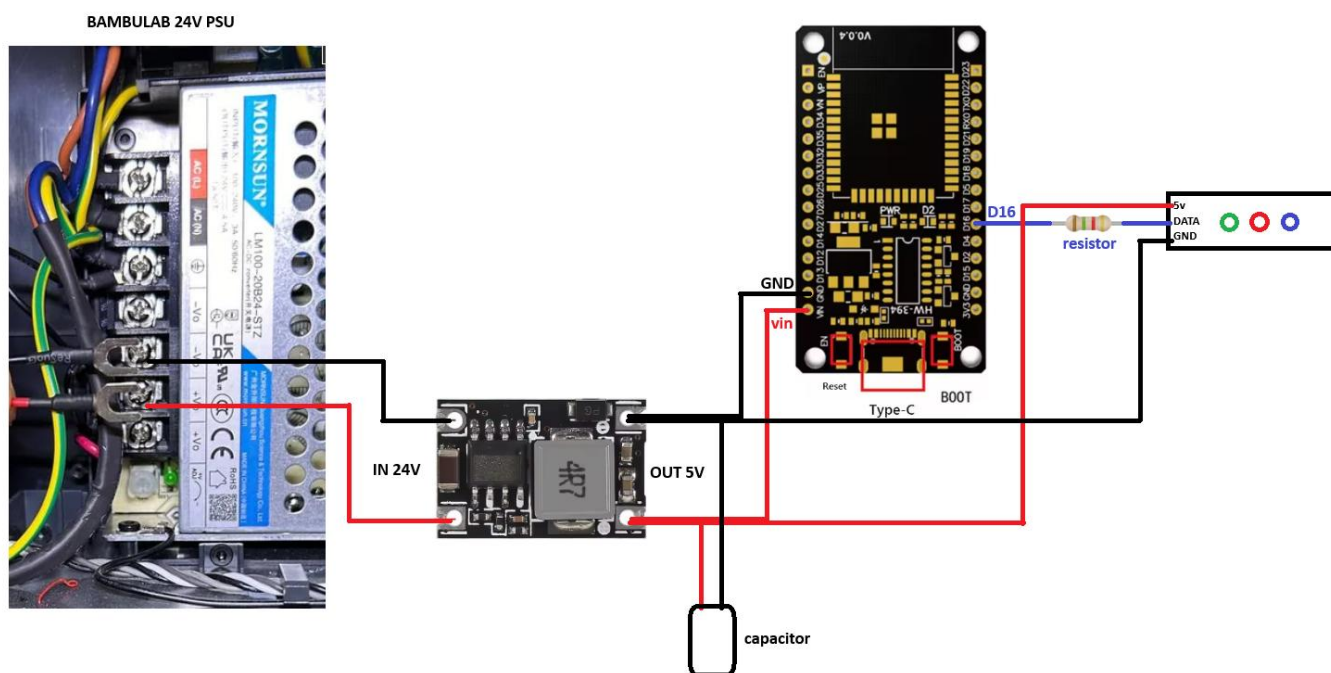
Wiring

The wiring is straightforward, we pull 24V from the printer power supply, step it down to 5V, power the ESP32 and the LED strip from that, and send one data wire from the ESP32 to the strip.

Two small extra parts help with stability:

470Ω resistor on the data wire

1000μF capacitor on the LED power input



1. Getting access inside the printer

Make sure the printer is fully powered off and unplugged before opening anything !!

To reach the wiring points we need, remove the back plate of the printer, and then take off the PSU cover to access the +24V and GND terminals.

It's just screws — nothing hidden.

If you want a visual guide for taking things apart, I recommend this video: [Remove poop chute](#) and [Remove PSU cover](#)

Once the PSU is exposed, you'll see the screw terminals where we connect power for the buck converter.

I used 3M double-sided tape to mount the buck converter and ESP32 directly onto the PSU cover, this keeps everything secure and out of the motion path. When the cover goes back on, the parts stay inside safely without touching anything else.

2. Running the LED strip cable cleanly + mounting the strip

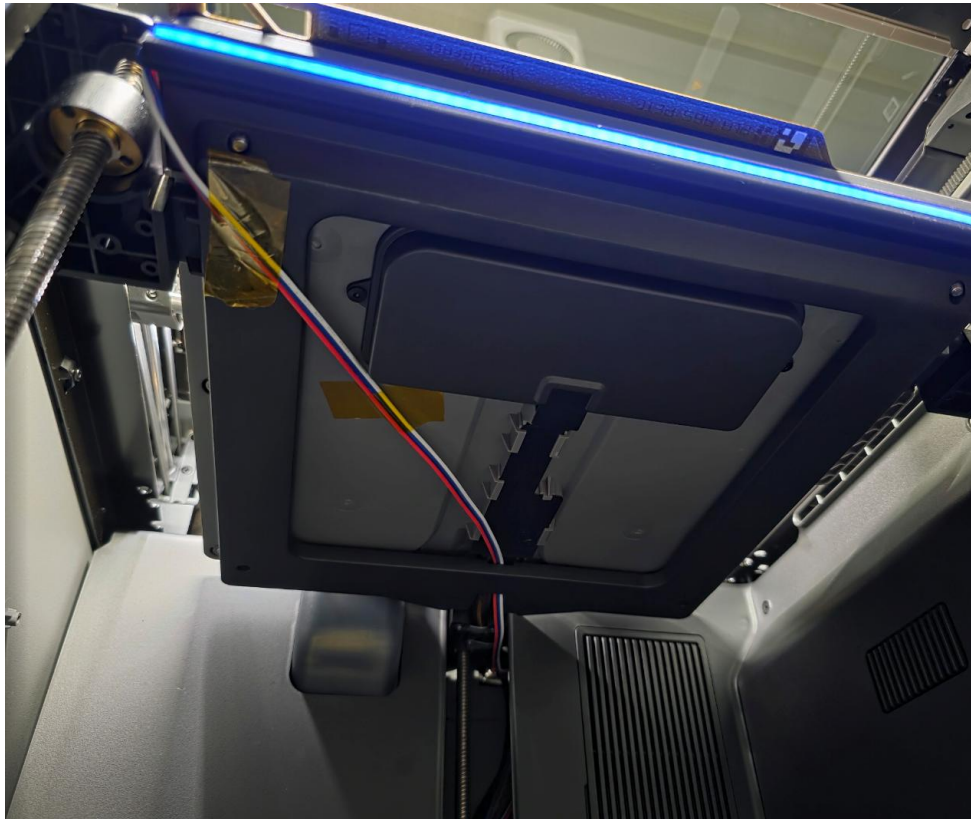
Print the LED strip mount.

For mounting the strip to the front of the bed frame, I used a 3D-printed LED strip holder (STL). The strip slides inside the printed mount, and then the mount itself gets attached to the front edge of the print-bed frame using 3M double-sided tape.



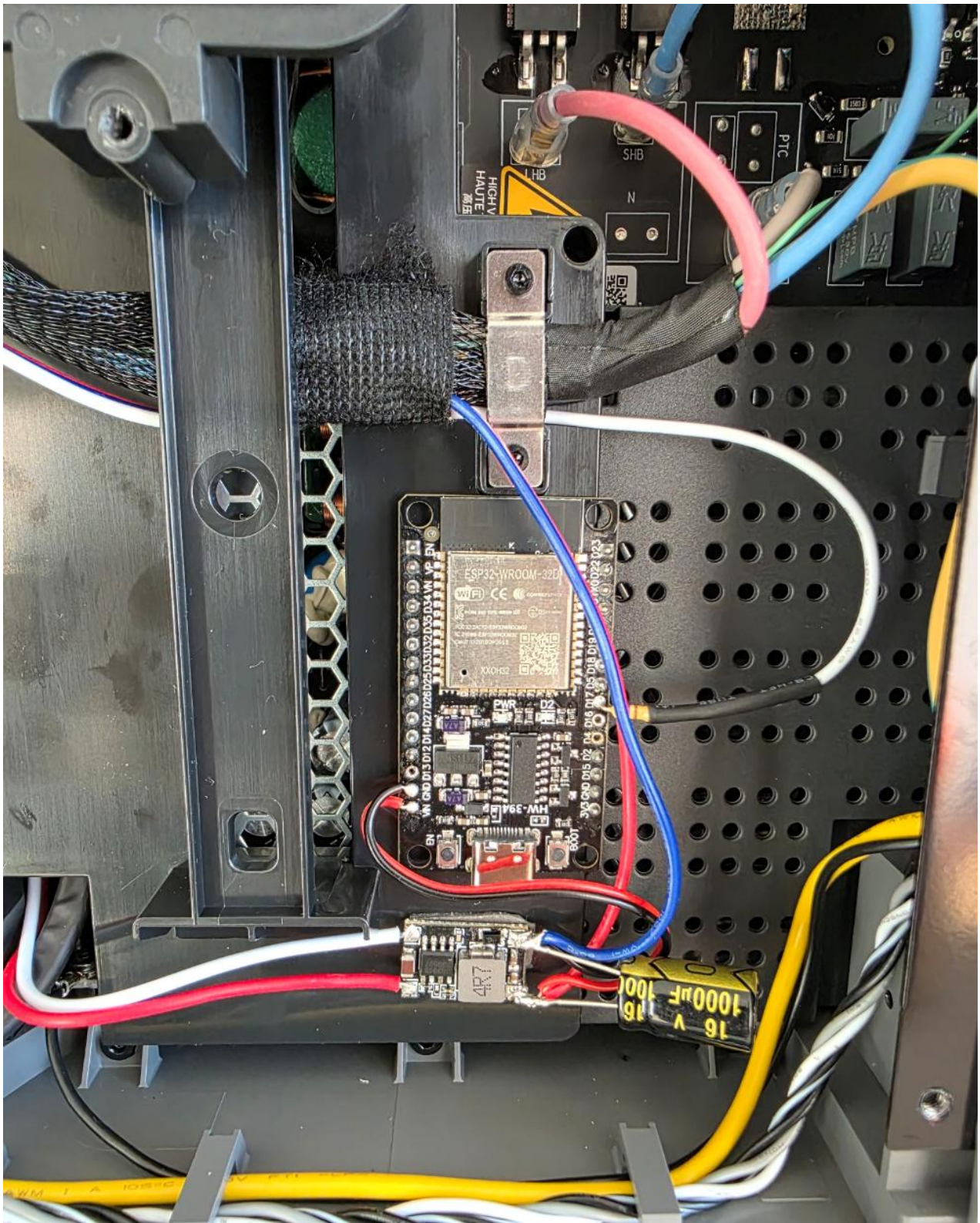
This keeps the strip protected, straight, and gives it a cleaner look than taping the LEDs directly to the frame.

- put 3M double-sided tape on the back of the printed mount
- stick the mount to the front edge of the bed frame, centered or wherever looks best to you
- run the LED strip cable along the existing bed wiring harness toward the back of the printer
- use zip ties or tape so the cable moves naturally with the bed and doesn't snag on anything



- feed the cable into the printer body and out the opening where the back panel was removed
- After everything is routed, connect the wiring to the buck converter, put the PSU cover back on, and close the rear plate.

The LED mount stays on the bed frame, the wiring follows the path of the original harness, and the ESP32 + buck live inside the printer mounted to the PSU cover.



Known Issue / Note:

When I power the printer through a smart outlet and turn it off, the LED strip sometimes flashes , even though everything should be off. I'm still looking into why it happens, feels like a small amount of leftover voltage or leakage somewhere, but not confirmed yet.

Project Links

For the latest version of the project, updates, files and community feedback, visit the MakerWorld project page:

Bambu Lab P2S LED Progress / Status Bar

[MakerWorld Project Page](#)

If you build this project, adapt it to another Bambu Lab printer, or improve the automation, feel free to share your results in the comments.

I'll keep updating the project page as I make improvements or get test results on other printers from other members.

CAD / Mount Source

You can also modify the LED holder using the original Onshape design:

[Onshape Document](#)

Thanks for checking out the project, and have fun building your own Bambu LED status bar. 😊